

Book 1 section 2.1: DSE multiple choice

2020 Q2

2. An electric kettle which contains 1 kg of water at room temperature takes 168 s to heat up the water to boiling point. The kettle's rated value is '220 V, 2000 W'. Assume that all the electrical energy consumed by the kettle is transferred to the water. Which of the following statements is/are correct?
Given: specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$
- (1) The initial temperature of the water is 20°C .
 - (2) The resistance of the kettle's heating element is about 24Ω .
 - (3) If the electric kettle is operated with 110 V, the time taken to heat up the water to boiling point will be doubled.
- A. (1) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (1), (2) and (3)